

SHASHWATA MANDAL

Ph.D. Candidate · Software Engineer

Ames, Iowa | 717-379-8394 | shashwata275@gmail.com
linkedin.com/in/shashwata-mandal | github.com/smoke275

SUMMARY

Ph.D. Candidate specializing in **Robotics and Autonomous Systems** with 6+ years of experience in algorithm design and full-stack implementation. Expert in bridging high-fidelity simulation (Gazebo, MuJoCo) with real-world embedded deployment (ROS2, PX4). Proven ability to deliver research-grade solutions for high-speed tracking and surveillance in complex, GPS-denied environments.

EDUCATION

Iowa State University <i>Ph.D. in Computer Science (GPA: 3.71)</i>	Ames, IA 2019 – May 2026 (<i>Expected</i>)
Pennsylvania State University <i>Master of Science in Computer Science (GPA: 3.47)</i>	Harrisburg, PA 2017 – 2019
West Bengal University of Technology <i>Bachelor of Technology in Computer Science and Engineering</i>	Kolkata, India 2012 – 2016

TECHNICAL SKILLS

Languages: C++ (Modern 14/17), Python, Java, Rust, JavaScript, HTML/CSS
Systems & Robotics: ROS/ROS2, Gazebo, MuJoCo, Isaac Sim/Lab, PX4 Autopilot, RViz, MavLink, CrazyFlie
Simulation & Viz: WebGL, Three.js, OpenCV, OpenGL, Physics Engine Integration
Hardware: Raspberry Pi 4/5, NVIDIA Jetson, Pixhawk, Lidar/Camera Integration
Dev Tools: Git, Docker, Linux (Ubuntu/Debian), CMake, CI/CD Pipelines

WORK EXPERIENCE

Research Intern (Path Planning) <i>Army Research Lab</i>	March 2025 – Present Adelphi, MD
- Engineered an entropy-based controller in a ROS-based simulation to maintain line-of-sight tracking, achieving target retention rates higher than state-of-the-art baselines. - Developed a Corner-Aware Risk Map module in Python to predict intruder escape routes, significantly reducing computation time for trajectory generation. - Designed a Neural Visibility Surrogate using supervised learning to approximate geometric oracle queries, reducing inference latency to 30ms for 1,000 points.	
Associate Software Engineer <i>Nomura Research Institute Financial Technologies Ltd.</i>	Aug 2016 – Aug 2017 Kolkata, India
- Led full-stack development of financial web platforms using Java Spring and JQuery, ensuring 99.9% uptime for critical trading modules. - Optimized backend SQL queries, reducing page load times by 30% during high-traffic Agile delivery phases.	

RESEARCH & PROJECTS

High-Speed Autonomous Pursuit System <i>Rust, JS, Python, ROS</i>	July 2022 – Present
ICRA 2025 Publication / ARL Collab: Architected a pursuit system capable of tracking high-velocity evaders in complex environments, outperforming state-of-the-art geometric methods. - Built a custom End-to-End Simulation Framework to support this research, featuring an HTML5/Canvas environment designer and a transpiler to auto-migrate maps into SDF/Gazebo. - Implemented a NeRF-based indoor caching architecture using Rust bindings to accelerate visual processing, enabling real-time lock-on for fast-moving targets. - Trained Corner-Aware RL agents within the framework, successfully handling severe occlusion where traditional heuristics failed.	
Battery Management System for Drone Fleets <i>Python, RL, Optimization</i>	Jan 2024 – Present
- Designed a drone-assisted BMS simulation for on-demand battery swaps, validating system viability for last-mile delivery networks. - Implemented Reinforcement Learning agents to optimize fleet efficiency, minimizing vehicle downtime and increasing daily delivery throughput.	
Multiple Target Tracking for Drone Swarms <i>ROS, PX4, C++</i>	Sep 2020 – May 2021
IROS 2021 Publication: Co-authored Planning for Aerial Robot Teams, introducing a minimum-time trajectory tracking algorithm. - Deployed full software stack on physical quadcopters (Pixhawk/PX4), conducting outdoor field tests to validate simulation results in real-world wind conditions.	